WholeStory Health

Multilingual Multimodal Clinical Communication Assistant

AI + Edge Hardware for Inclusive Healthcare

Cornell Engineers Without Borders – Digital Agriculture

THE PROBLEM

1 in 5

U.S. patients speaks a language other than English at home

36%

of adults have low health literacy — struggle with medical terminology

10–30 min

patients wait before seeing a doctor — time currently wasted

Result: Doctors miss critical context. Patients leave confused. Documentation suffers.

A REAL VISIT — WITHOUT OUR SYSTEM

1

Patient Arrives

Speaks Wolof. Fills paper form in a language they don't fully read.

2

Waits 20 Minutes

Alone in the exam room. No intake. No communication. Lost time.

3

Doctor Enters

Has 10 minutes. Asks rushed questions. Translation is incomplete.

4

Patient Leaves Confused

Didn't fully understand the diagnosis or instructions. No follow-up clarity.

WholeStory Health solves this — before, during, and after the visit.

OUR SOLUTION

WholeStory Health

A multilingual AI-powered clinical assistant that helps patients communicate symptoms and helps doctors start every visit with a clearer, fuller understanding of the patient.



Pre-Visit Intake

AI companion talks to the patient during the wait — collects symptoms, history, life context



Live Translation

Real-time multilingual speech translation with dialect awareness during the visit



Visual Interface

Body maps, pain scale faces, symptom illustrations — works across languages & literacy levels



Clinical Notes

Auto-generates structured doctor summary and clinical documentation from the encounter

VISUAL SYMPTOM INTERFACE

Body Map

Patient taps exactly where it hurts on a human body diagram. System zooms in and asks follow-up questions.

Pain Scale Faces

Smiley-to-distressed face scale. No reading required — universally understood across languages.

"What Do You Mean?" Button

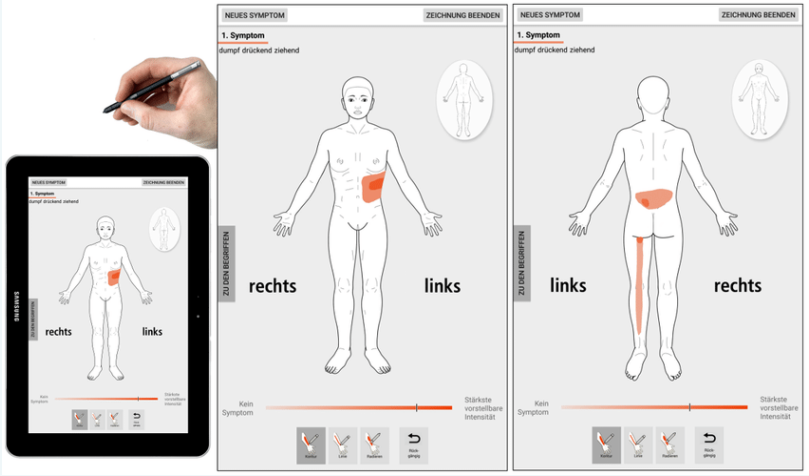
If doctor uses a complex term, patient presses this. System shows an image, simple explanation, and audio in their language.

Doctor Explanation Mode

Doctor says 'airway inflammation' — system shows a lung diagram, highlights the area, and plays a simplified audio explanation.

WholeStory Health

Where does it hurt?



MULTILINGUAL AI CONVERSATION

⚠ No system today reliably translates dialects in real medical conversations. This is one of the biggest unsolved problems in healthcare AI.



Supported Languages & Dialects (current + target):

English	French	Arabic	Spanish	Wolof
Amharic	Swahili	Hausa	+ more	

Why this matters: Minority languages & dialects are severely underrepresented in medical AI training data.

EDGE AI HARDWARE SYSTEM

DEVICE 1 — Clinic Kiosk / Exam Room

- Beamforming microphone array
- Touchscreen display (multilingual UI)
- NVIDIA Jetson Orin Nano (edge AI)
- Encrypted local storage
- Secure hospital network option
- Privacy indicator light

feeds

DEVICE 2 — Global Dialect Collection Pod

- Solar + 6-month battery life
- LoRa / cellular connectivity
- Low-cost MEMS microphone
- SD card local storage
- Consent-first guided prompts
- Auto-anonymization pipeline

ACCESSIBILITY & EMOTION-AWARE AI



Emotion & Stress Detection

- Detects voice stress, distress, hesitation
- Flags confusion or possible urgent concern
- Helps doctor know patient may need extra time
- Not a diagnosis — a supportive signal



Low-Literacy Interface

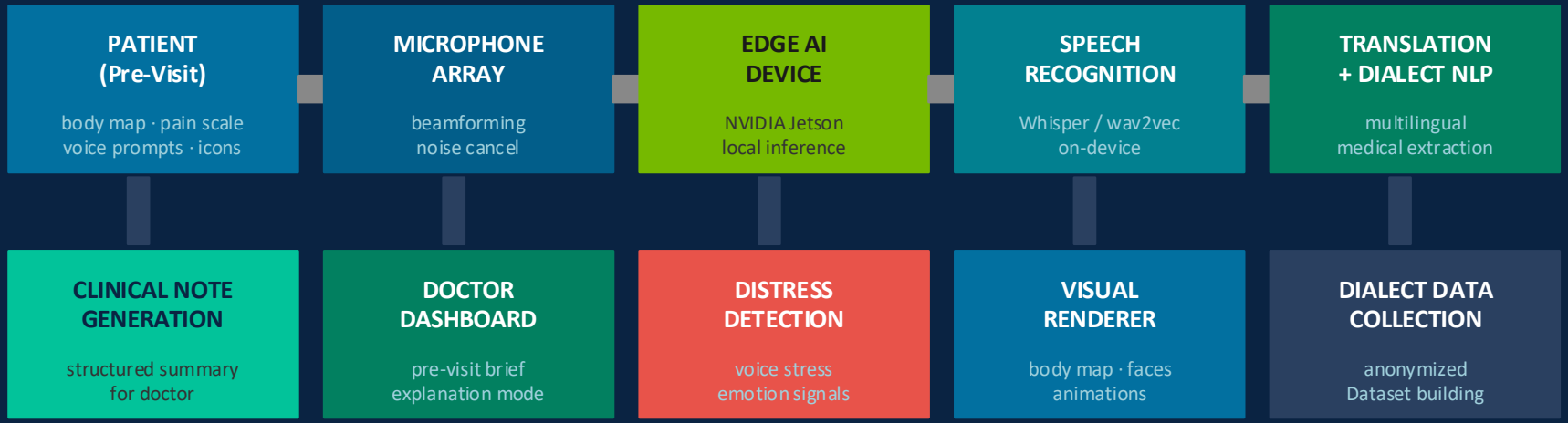
- Full voice prompt navigation
- Big icons & illustrated buttons
- No text required to use the system
- Yes/No physical buttons available



Fully Offline AI

- All processing runs on device
- Works in rural & low-connectivity clinics
- Patient data never leaves the room
- Sync to cloud when connected

SYSTEM ARCHITECTURE



All computation runs locally on the edge device — no cloud required for core functionality

Compute	Input	Storage	Power
NVIDIA Jetson Orin Nano NVIDIA Jetson Xavier NX	Beamforming mic array Touchscreen (multilingual)	AES-256 encrypted SD Optional HIPAA cloud sync	Solar + 6-month battery (global collection pod)

GLOBAL IMPACT



7,000+

Languages & Dialects Worldwide



800M+

Adults with Low Health Literacy



68M

Refugees & Displaced Persons (UNHCR 2023)



4.5B

People Lack Basic Healthcare Access

Who Benefits:

- ✓ Hospitals & Clinics
- ✓ Rural Health Programs
- ✓ Refugee & Immigrant Communities
- ✓ Global Health NGOs
- ✓ Medical AI Researchers
- ✓ Language & Accessibility Advocates

WholeStory Health

Every patient deserves to be heard — in any language,
at any literacy level, anywhere in the world.

Cornell University — Engineers Without Borders Digital Agriculture

https://ewb-dig-ag.org/